Project Development Phase

Data Preprocessing Steps and Business Questions with Visualization

Date	06 July 2026
Team ID	
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

1.Preprocessing Steps

The raw UNESCO World Heritage Sites dataset was cleaned and prepared in Tableau / Excel before any visualizations were built. The key steps carried out were:

- **Gathering Data — downloaded the raw UNESCO World Heritage Sites dataset (CSV) from the open data source.**
- **Loading Data — imported the dataset into Tableau Desktop for analysis.**
- **Handling Missing Values — identified and cleaned null/blank fields such as Danger status, Area Hectares, and Date Inscribed/Date End.**
- **Creating Fields — derived calculated fields including Category (Cultural/Mixed/Natural), Region groupings, and a cleaned Danger indicator (0/1).**
- **Handling Inconsistency in Data — standardized country and region name spellings, and unified date formats so YEAR() calculations work correctly across the dataset.**

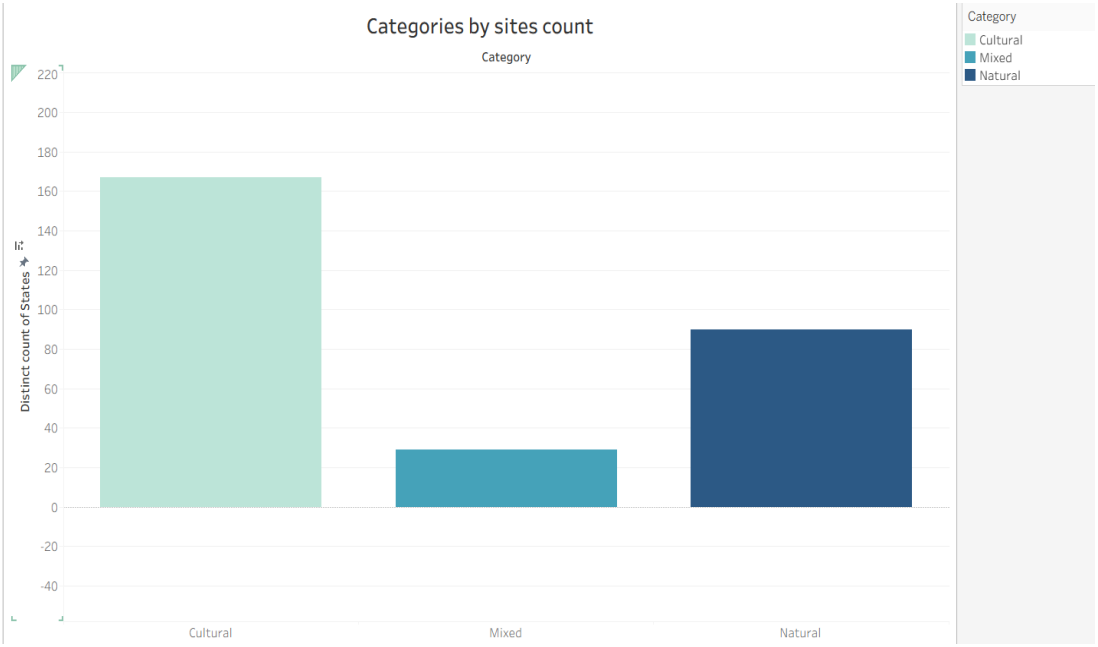
These steps ensured the dataset used consistent field names, valid geographic roles (for Country/Region mapping), and numeric/boolean formats suitable for aggregation (COUNT, CNTD, SUM) across all worksheets.

2. Business Questions, Answers and Visualizations

Each chart in the dashboard was built to answer a specific business question about UNESCO World Heritage Sites. The question, the data-driven answer, and the corresponding chart are shown below.

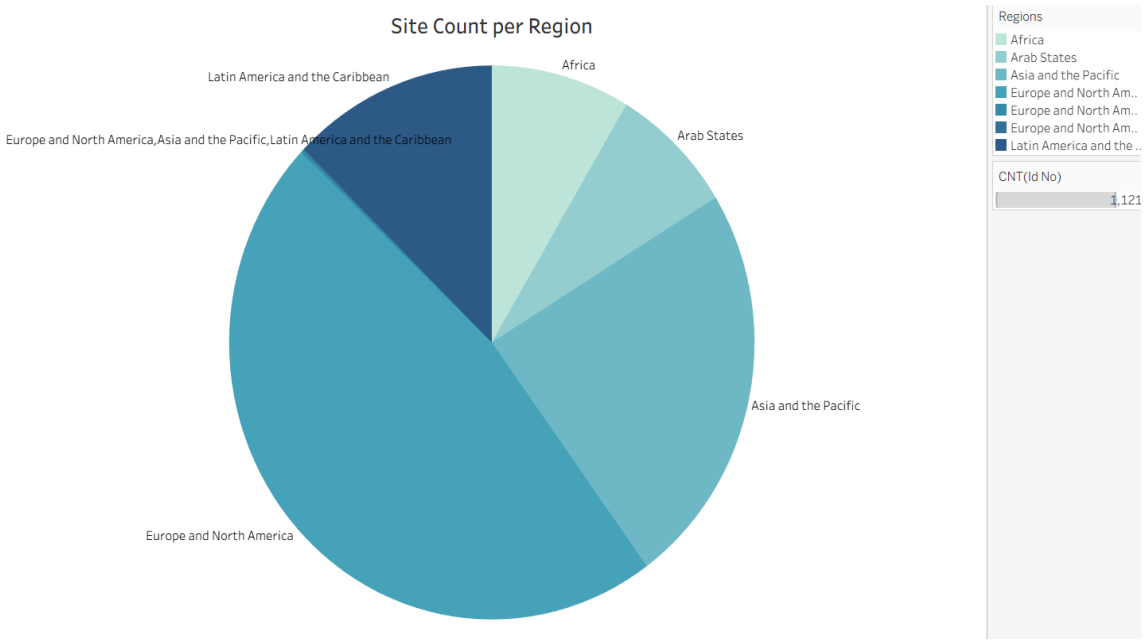
Q1. Which category of heritage sites is most common worldwide?

Answer: Cultural sites make up the largest share of all UNESCO World Heritage Sites, followed by Natural sites, while Mixed sites (both cultural and natural significance) are the least common category.



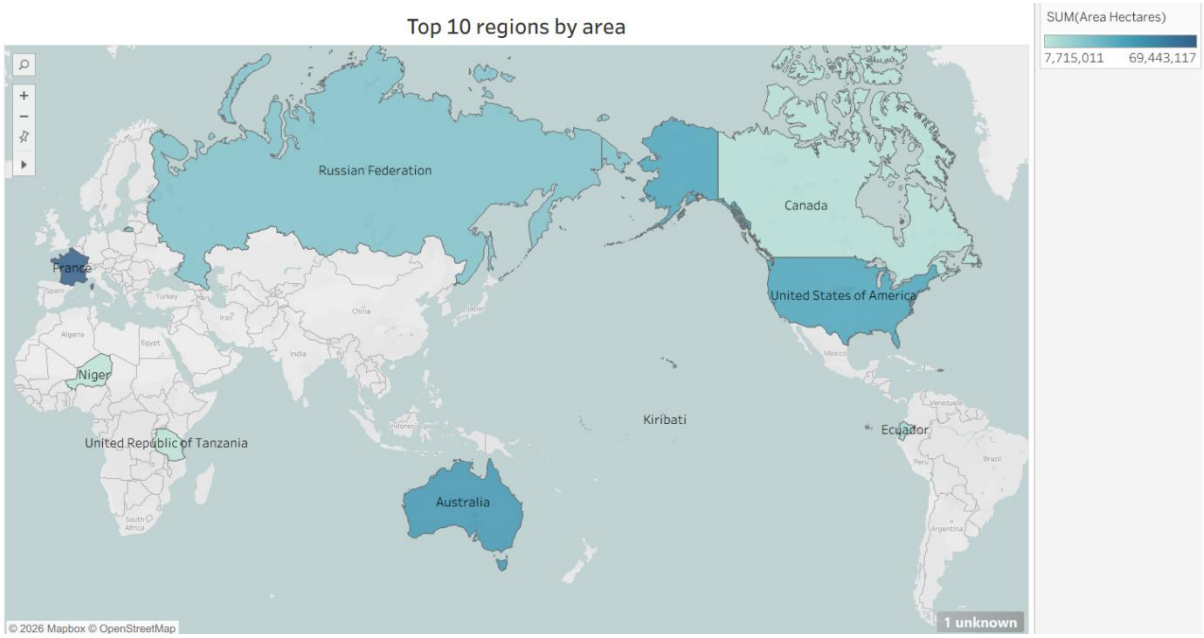
Q2. How are heritage sites distributed across world regions?

Answer: Europe and North America and Asia and the Pacific together account for the largest share of heritage sites, while the Arab States and Latin America and the Caribbean hold smaller shares.



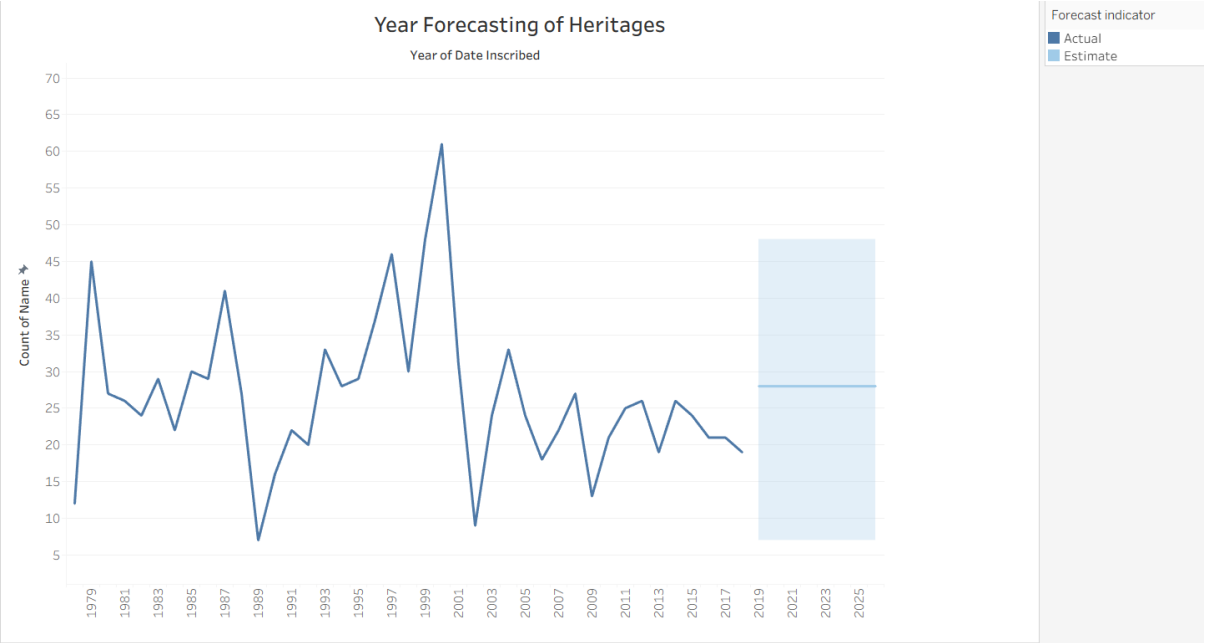
Q3. Which countries/regions hold the largest heritage sites by area?

Answer: The Russian Federation, Canada, and the United States of America appear among the countries with the largest heritage site area, alongside France, Niger, the United Republic of Tanzania, Kiribati, Ecuador and Australia in the top 10.



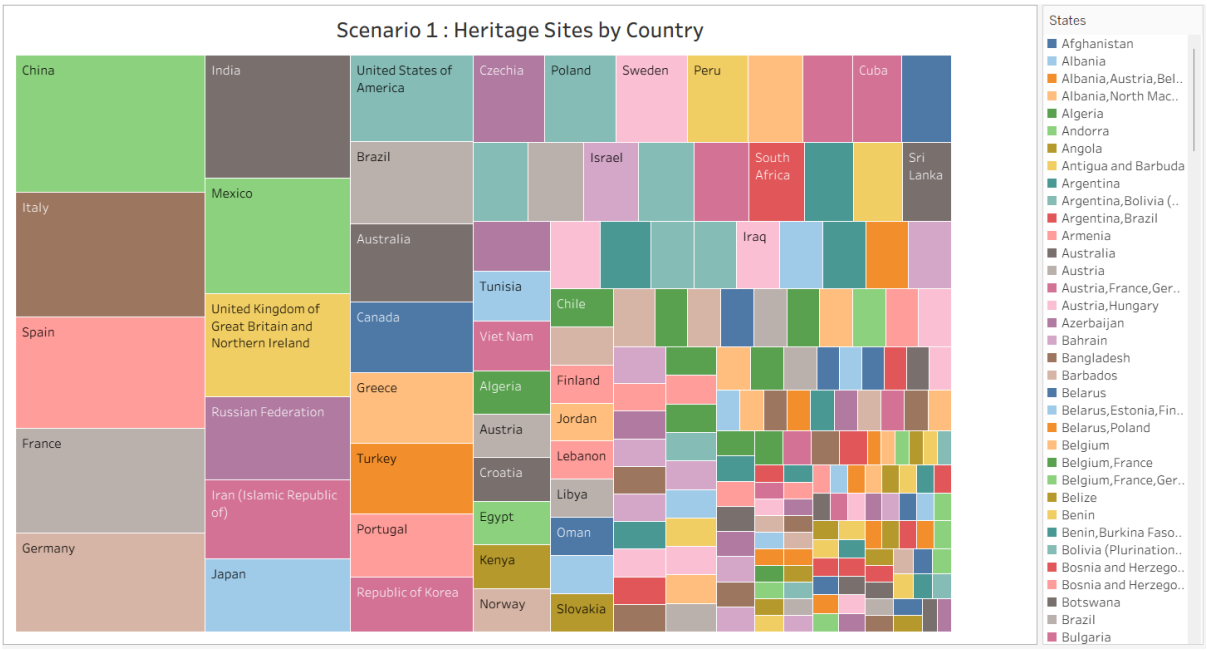
Q4. How has the number of new heritage site inscriptions changed over the years, and what is the forecast?

Answer: Inscriptions rose steadily from the late 1970s, peaking around 1999-2000 with roughly 60 new sites in a single year. Since then, yearly inscriptions have fluctuated between 10 and 40, and Tableau's built-in forecast projects this moderate, fluctuating pace to continue.



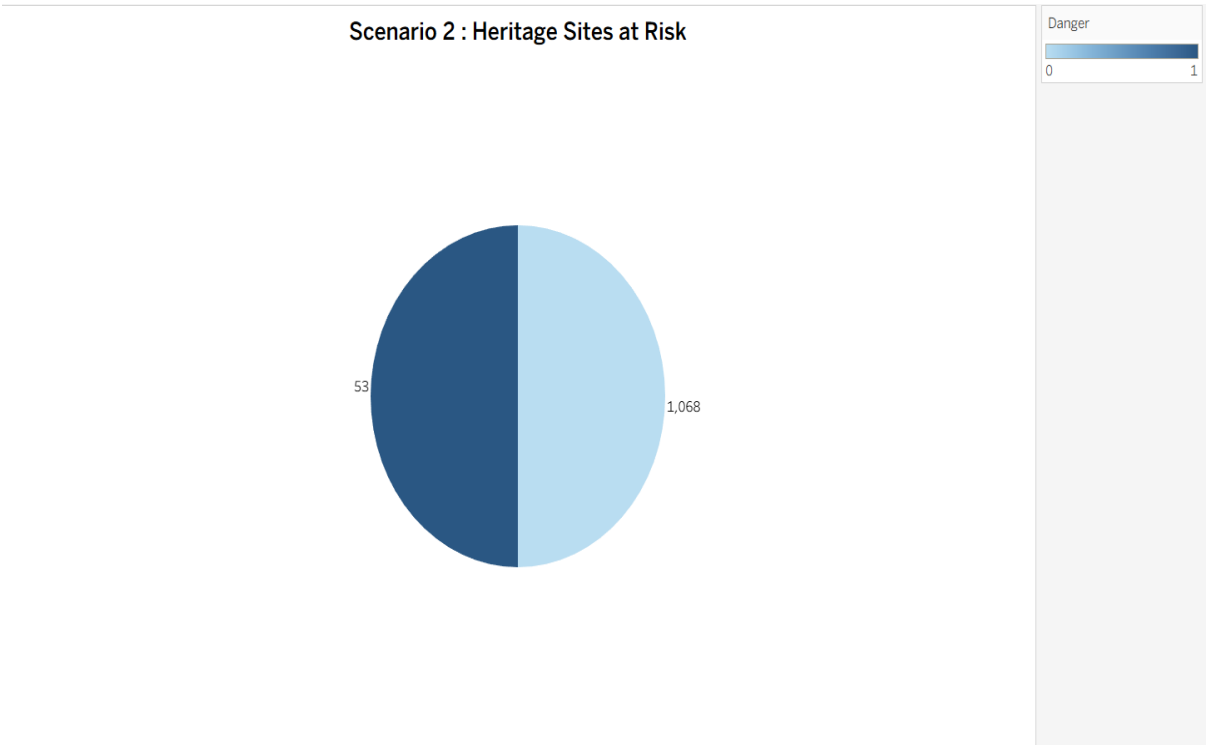
Q5. Which countries have the highest number of heritage sites?

Answer: China, India, Italy, Spain and France stand out as the countries with the largest number of heritage sites, shown as the largest blocks in the treemap.



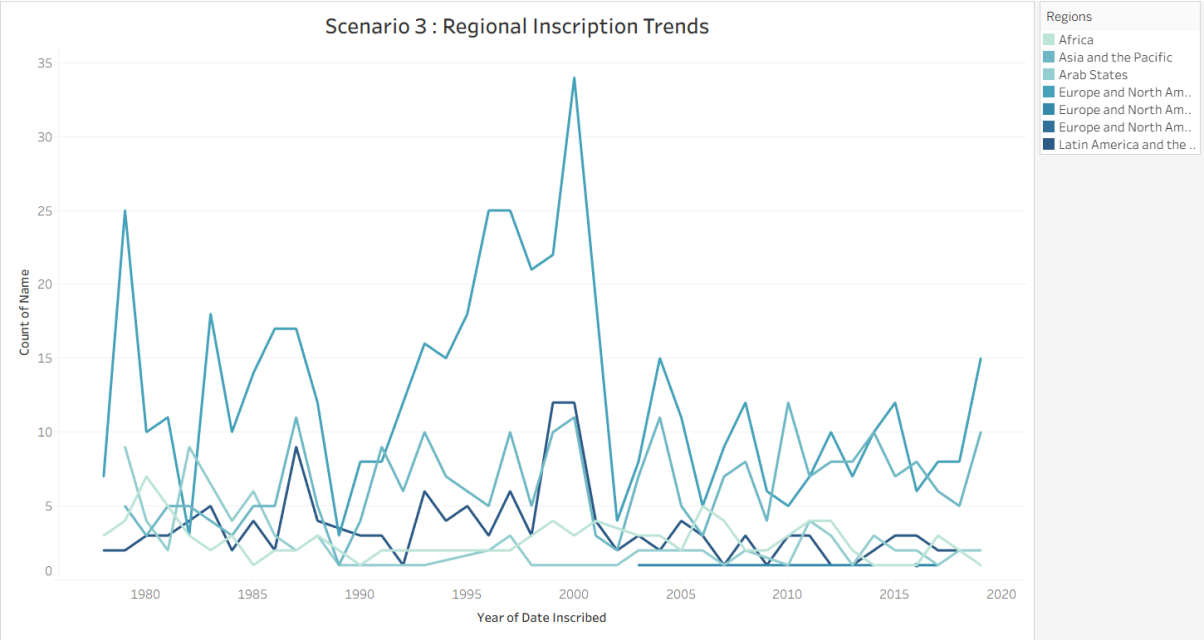
Q6. How many countries fall under each region?

Answer: Europe and North America has the most countries (71) represented, followed by Asia and the Pacific (38) and Africa (37), while Latin America and the Caribbean and the Arab States (19) have fewer.



Q8. Which regions show the strongest growth trends in heritage site inscriptions over time?

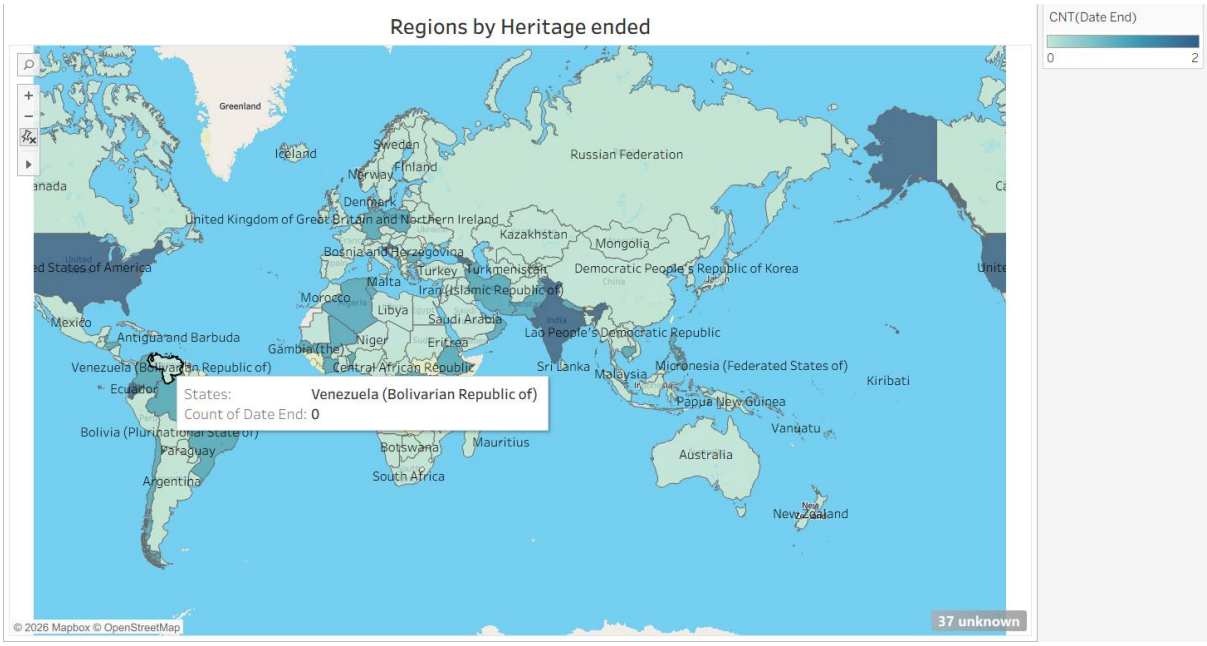
Answer: Europe and North America shows the most pronounced spike in inscriptions, with a sharp peak around the year 2000, while other regions show steadier, more gradual growth trends over the same period



Additional Supporting Visualizations (Dashboard 2)

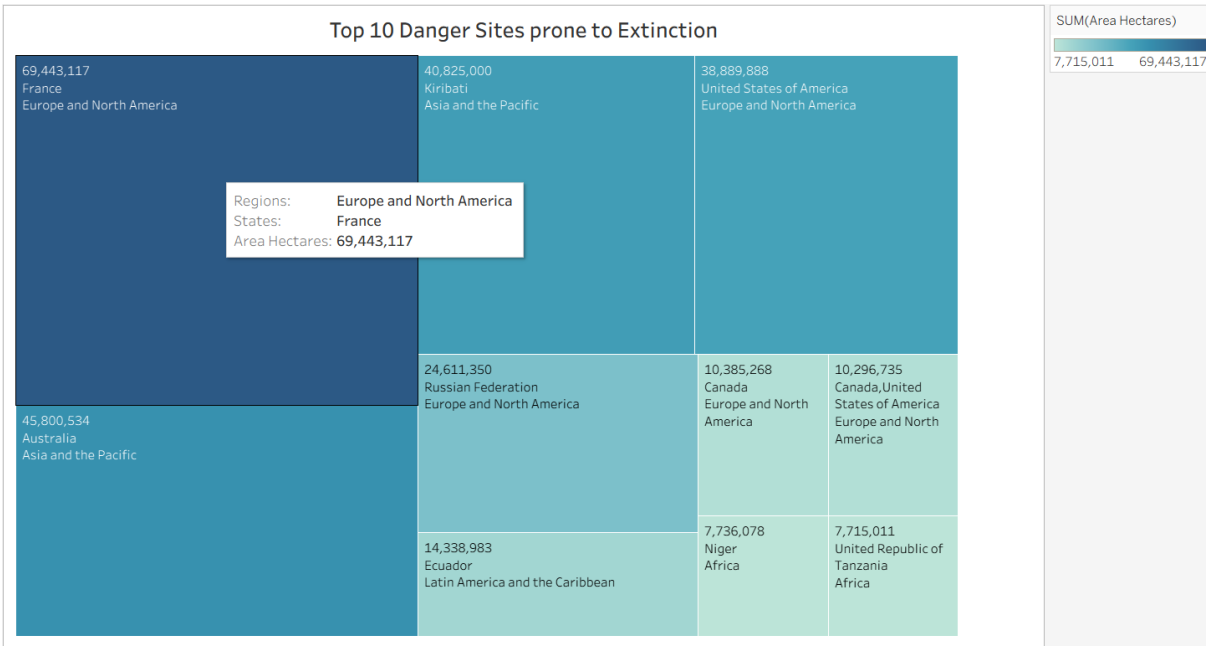
Regions by Heritage Ended

Highlights countries where a heritage site's status has ended (delisted), including Iceland, the United Kingdom, Iran, Thailand, Venezuela, Papua New Guinea, Bolivia, Australia and New Zealand



Top 10 Danger Sites prone to Extinction

Ranks the largest at-risk sites by area, led by a site in France (~69.4M hectares), Australia (~45.8M hectares) and Kiribati (~40.8M hectares).



Analysis of Danger Sites and Area of Heritages

Compares the count of danger-flagged sites (bars) against total area in hectares (line) by year, highlighting a large area spike around 2009.

